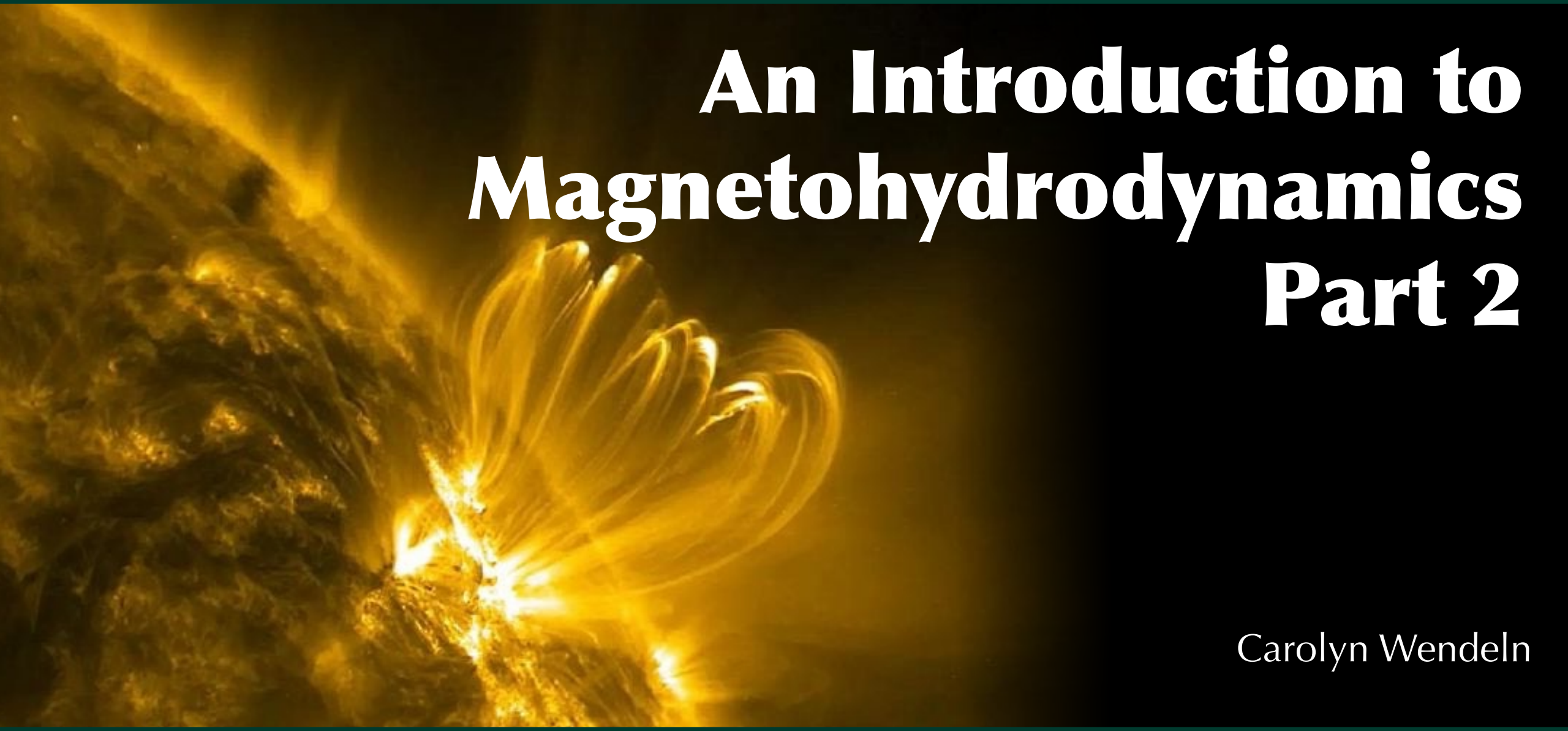

# An Introduction to Magnetohydrodynamics Part 2

Carolyn Wendeln

# Resources

- Fundamentals of Plasma Physics by J. A. Bittencourt
- Introduction to Plasma Physics and Controlled Fusion by Francis F. Chen
- Physics of the Space Environment by Tamas I. Gombosi
- Finite Volume Methods for Hyperbolic Problems by Randall J. LeVeque
- Numerical Methods for Conservation Laws by Randall J. LeVeque
- Riemann Solvers and Numerical Methods for Fluid Dynamics by Eleuterio F. Toro
- Introduction to Computational Astrophysical Hydrodynamics by Michael Zingale

# The Ideal MHD Equations

$$\partial_t \begin{bmatrix} \rho \\ \rho\mathbf{u} \\ \mathcal{E} \\ \mathbf{B} \end{bmatrix} + \nabla \cdot \begin{bmatrix} \rho\mathbf{u} \\ \rho\mathbf{u} \otimes \mathbf{u} + \left(p + \frac{1}{2}\|\mathbf{B}\|^2\right)\mathbb{I} - \mathbf{B} \otimes \mathbf{B} \\ \mathbf{u}\left(\mathcal{E} + p + \frac{1}{2}\|\mathbf{B}\|^2\right) - \mathbf{B}(\mathbf{u} \cdot \mathbf{B}) \\ \mathbf{u} \otimes \mathbf{B} - \mathbf{B} \otimes \mathbf{u} \end{bmatrix} = \mathbf{0} \tag{1}$$

$$\nabla \cdot \mathbf{B} = 0 \tag{2}$$

$$\mathcal{E} = \frac{p}{\gamma - 1} + \frac{1}{2}\rho\|\mathbf{u}\|^2 + \frac{1}{2}\|\mathbf{B}\|^2 \tag{3}$$

$\rho$: Density
$\mathbf{u}$: Velocity
$\mathcal{E}$: Total Energy
$\mathbf{B}$: Magnetic Field
$p$: Pressure
$\gamma$: Ideal Gas Constant

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$$\boxed{\nabla \cdot \mathbf{B} = 0} \tag{2}$$

Part II

$$\mathcal{E} = \frac{p}{\gamma - 1} + \frac{1}{2}\rho\|\mathbf{u}\|^2 + \frac{1}{2}\|\mathbf{B}\|^2 \tag{3}$$

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# Outline

- Paper 0: Tóth An Overview of Methods for the $\nabla \cdot \mathbf{B} = 0$ Constraint
- Paper 1: Powell 8-Wave Formulation
- Paper 2: Evans & Hawley Constrained Transport
- Paper 3: Rossmanith Unstaggered Constrained Transport
- Paper 4: Brackbill & Barnes Projection Method
- Paper 5: Dedner et. al. Hyperbolic Divergence Cleaning

# Paper #0: Great Review of Different Methods

The $\nabla \cdot \mathbf{B} = 0$ Constraint in Shock-Capturing Magnetohydrodynamics Codes

Gábor Tóth - 2000

# Paper #1: The 8-Wave Formulation

An Approximate Riemann Solver for Magnetohydrodynamics
(That Works In More Than One Dimension)

Kenneth G. Powell - 1994

# Paper #1: The 8-Wave Formulation

- The earliest MHD Riemann solvers only account for **seven** waves
- e.g., Brio & Wu, Dai & Woodward

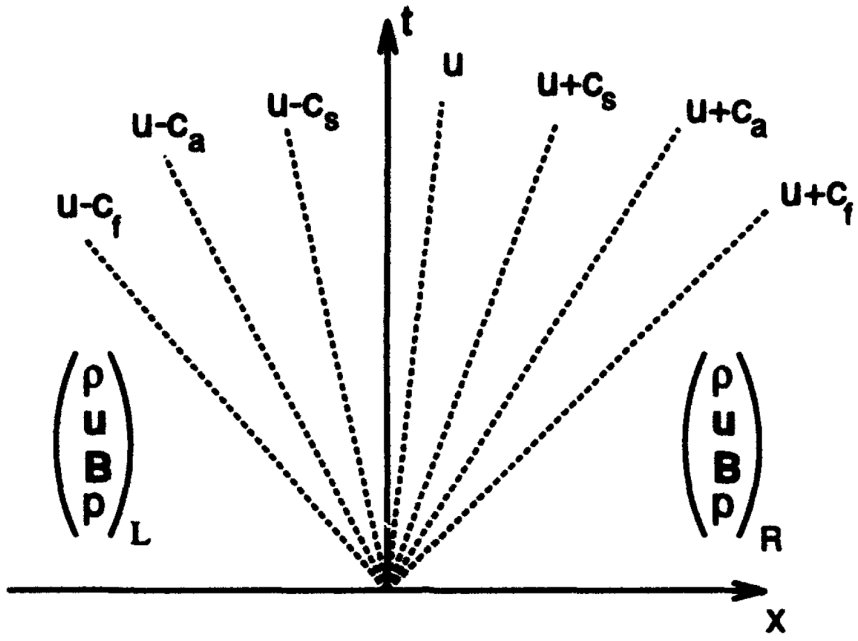


Figure 1: Waves in the One-Dimensional MHD Riemann Problem

# Paper #1: The 8-Wave Formulation

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- 1D: $B_x$ is constant

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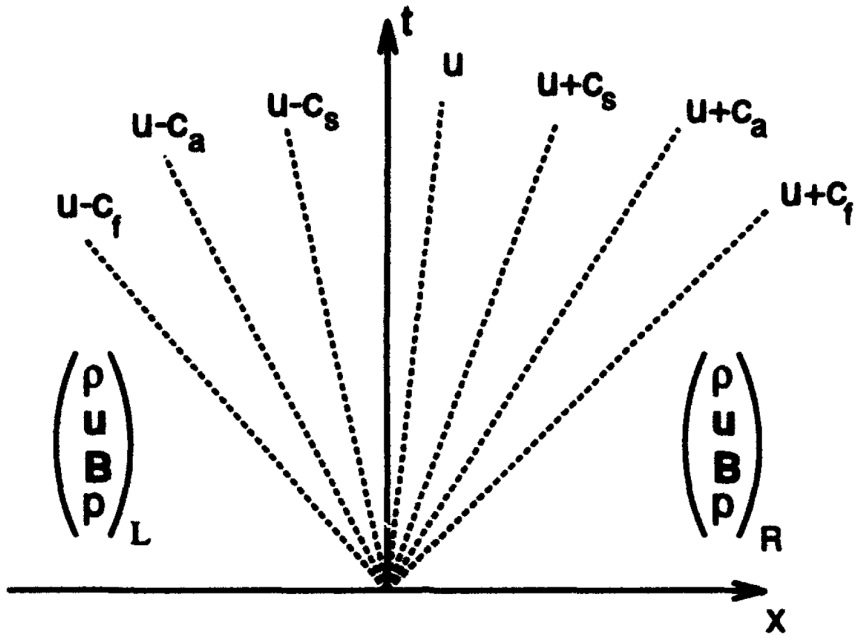


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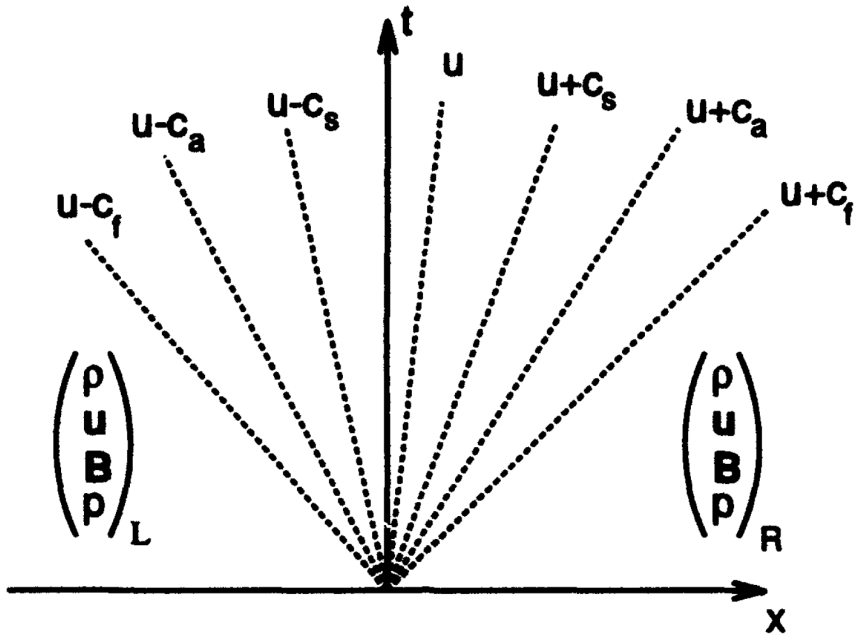


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- 2D/3D: $B_x$ can change

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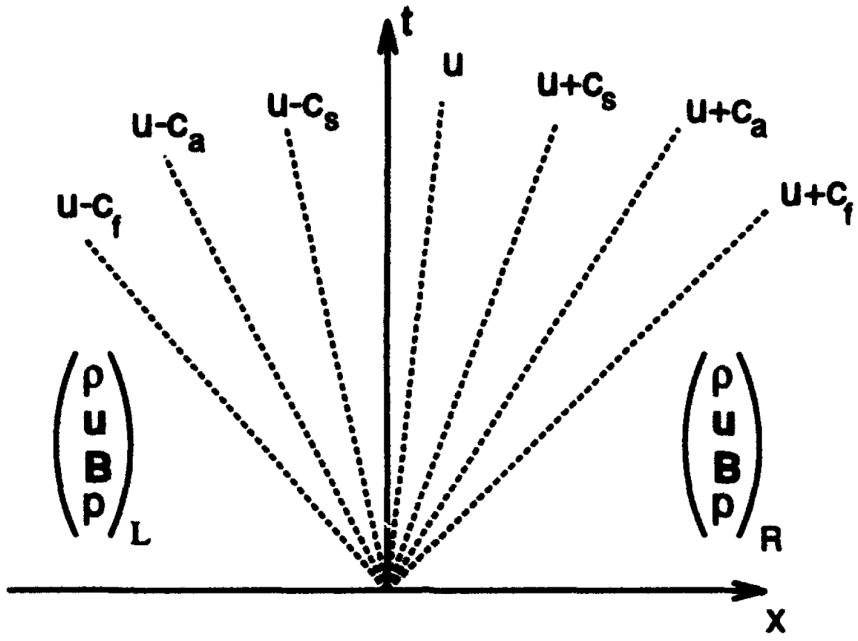


Figure 1: Waves in the One-Dimensional MHD Riemann Problem

# Paper #1: The 8-Wave Formulation

- Given the primitive variables: $\mathbf{W} = (\rho, u, v, w, B_x, B_y, B_z, p)$

- The Jacobian of the system is given as:

$$A_p = \begin{bmatrix} u & \rho & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & u & 0 & 0 & -\frac{B_x}{\rho} & \frac{B_y}{\rho} & \frac{B_z}{\rho} & \frac{1}{\rho} \\ 0 & 0 & u & 0 & -\frac{B_y}{\rho} & -\frac{B_x}{\rho} & 0 & 0 \\ 0 & 0 & 0 & u & -\frac{B_z}{\rho} & 0 & -\frac{B_x}{\rho} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & B_y & -B_x & 0 & -v & u & 0 & 0 \\ 0 & B_z & 0 & -B_x & -w & 0 & u & 0 \\ 0 & \gamma p & 0 & 0 & (\gamma - 1)\,\mathbf{u} \cdot \mathbf{B} & 0 & 0 & u \end{bmatrix}$$

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**Bad!**

# Paper #1: The 8-Wave Formulation

- The matrix is singular (not invertible)
- It leads to a non-physical zero eigenvalue

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- This paper derives an 8-Wave Riemann Solver
- It modifies $A_p$ to become a non-singular modified matrix $A_p^{'}$
- The criteria of $A_p^{'}$ are…

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- The eigenvalues and eigenvectors corresponding to the 7-waves in the 1D Riemann Solver must remain unchanged
- The eigenvalue corresponding to the new eighth wave is equal to u
- The left and right eigenvectors corresponding to the new eight wave "make sense”
- In the case where $B_x$ = constant, the 8-wave Riemann problem reduces to the 7-wave Riemann problem

# Paper #1: The 8-Wave Formulation

Given some patience & facility with Maple's symbolic manipulation capabilities …

$$A_p = \begin{bmatrix} u & \rho & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & u & 0 & 0 & -\frac{B_x}{\rho} & \frac{B_y}{\rho} & \frac{B_z}{\rho} & \frac{1}{\rho} \\ 0 & 0 & u & 0 & -\frac{B_y}{\rho} & -\frac{B_x}{\rho} & 0 & 0 \\ 0 & 0 & 0 & u & -\frac{B_z}{\rho} & 0 & -\frac{B_x}{\rho} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & B_y & -B_x & 0 & -v & u & 0 & 0 \\ 0 & B_z & 0 & -B_x & -w & 0 & u & 0 \\ 0 & \gamma p & 0 & 0 & (\gamma-1)\,\mathbf{u}\cdot\mathbf{B} & 0 & 0 & u \end{bmatrix}$$

$$A'_p = \begin{bmatrix} u & \rho & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & u & 0 & 0 & 0 & \frac{B_y}{\rho} & \frac{B_z}{\rho} & \frac{1}{\rho} \\ 0 & 0 & u & 0 & 0 & -\frac{B_x}{\rho} & 0 & 0 \\ 0 & 0 & 0 & u & 0 & 0 & -\frac{B_x}{\rho} & 0 \\ 0 & 0 & 0 & 0 & u & 0 & 0 & 0 \\ 0 & B_y & -B_x & 0 & 0 & u & 0 & 0 \\ 0 & B_z & 0 & -B_x & 0 & 0 & u & 0 \\ 0 & \gamma p & 0 & 0 & 0 & 0 & 0 & u \end{bmatrix}$$

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# Paper #1: The 8-Wave Formulation

Question: How has $A_p'$ modified our system of conservation laws?

$$\frac{\partial}{\partial t}\begin{pmatrix} \rho \\ \rho\mathbf{u} \\ \mathbf{B} \\ E \end{pmatrix} + \nabla\cdot\begin{pmatrix} \rho\mathbf{u} \\ \rho\mathbf{u}\mathbf{u} + \mathbf{I}\left(p + \frac{\mathbf{B}\cdot\mathbf{B}}{2}\right) - \mathbf{B}n\mathbf{B} \\ \mathbf{u}\mathbf{B} - \mathbf{B}\mathbf{u} \\ \left(E + p + \frac{\mathbf{B}\cdot\mathbf{B}}{2}\right)\mathbf{u} - \mathbf{B}\left(\mathbf{u}\cdot\mathbf{B}\right) \end{pmatrix} = 0$$

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Added source terms that are proportional to $\nabla\cdot\mathbf{B} = 0$

# Paper #1: The 8-Wave Formulation

- Pros:
  - The deviations from conservation will be very small
  - Improved robustness and accuracy compared to the 7-wave formulation
- Cons:
  - Does not ensure $\nabla \cdot \mathbf{B} = 0$
  - Our system is no longer conservative
  - The non-conservative source terms can produce incorrect shock speeds

# Paper #2: Constrained Transport

Simulation of Magnetohydrodynamic Flows:
A Constrained Transport Method

Charles R. Evans & John F. Hawley - 1988

# Paper #2: Constrained Transport

- Because it is divergence-free, the magnetic field $\mathbf{B}$ can be written as the curl of a magnetic vector potential

$$\mathbf{B} = \nabla \times \mathbf{A}$$

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- Because it is divergence-free, the magnetic field $\mathbf{B}$ can be written as the curl of a magnetic vector potential

$$\mathbf{B} = \nabla \times \mathbf{A}$$

- *What does that mean?*

# Paper #2: Constrained Transport

- **Helmholtz Decomposition Theorem**: A vector field $\mathbf{F}$ can be written as the gradient of a scalar $\phi$ plus the curl of a vector $\mathbf{A}$

$$\mathbf{F} = \nabla\phi + \nabla \times \mathbf{A}$$

# Paper #2: Constrained Transport

- **Helmholtz Decomposition Theorem**: A vector field $\mathbf{F}$ can be written as the gradient of a scalar $\phi$ plus the curl of a vector $\mathbf{A}$

$$\mathbf{F} = \nabla\phi + \nabla \times \mathbf{A}$$

- If $\mathbf{B} = \nabla \times \mathbf{A}$, then $\nabla \cdot \mathbf{B} = \nabla \cdot (\nabla \times \mathbf{A}) = 0$ is always true

# Paper #2: Constrained Transport

- Because it is divergence-free, the magnetic field $\mathbf{B}$ can be written as the curl of a magnetic vector potential $\mathbf{A}$

$$\mathbf{B} = \nabla \times \mathbf{A}$$

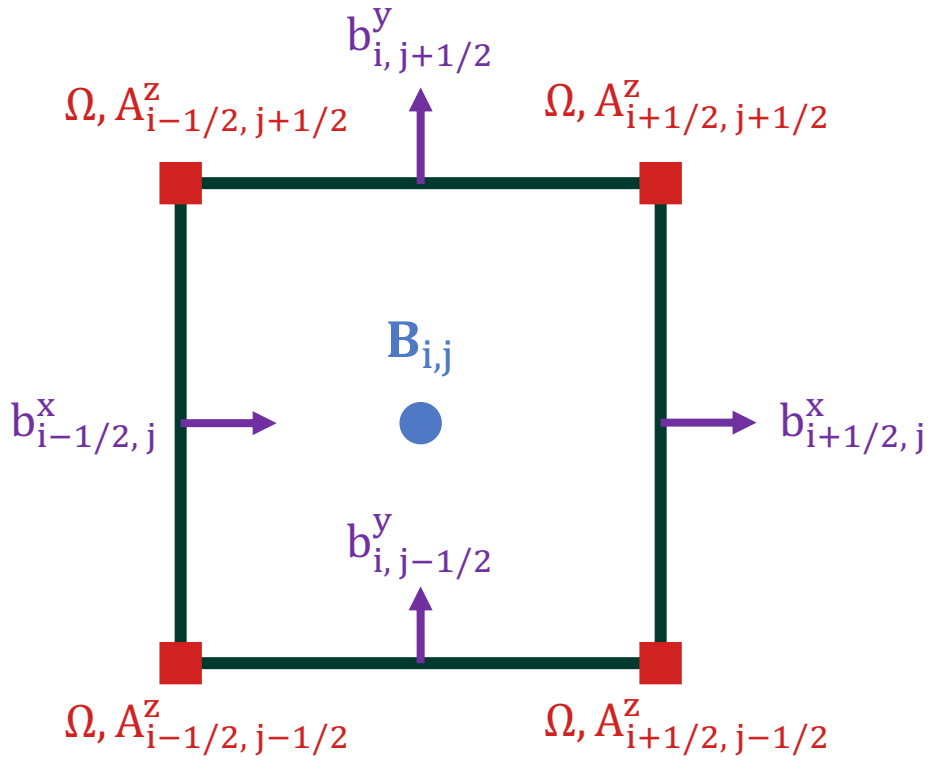
- The Induction equation can be rewritten as

$$\frac{\partial \mathbf{A}}{\partial \mathrm{t}} + \mathbf{\Omega} = 0$$

where $\mathbf{\Omega} = \mathbf{B} \times \mathbf{u}$ is the electric field (from Ohm's law)

# Paper #2: Constrained Transport

- At cell centers: $\rho_{i,j}$, $\rho\mathbf{u}_{i,j}$, $\mathcal{E}_{i,j}$, $\mathbf{B}_{i,j}$
- At cell corners: $\Omega^z_{i-1/2,\,j-1/2}$, $A^z_{i-1/2,\,j-1/2}$, …
- At left/right edges: $b^x_{i-1/2,\,j}$, $b^x_{i+1/2,\,j}$
- At top/bottom edges: $b^y_{i,\,j-1/2}$, $b^y_{i,\,j+1/2}$



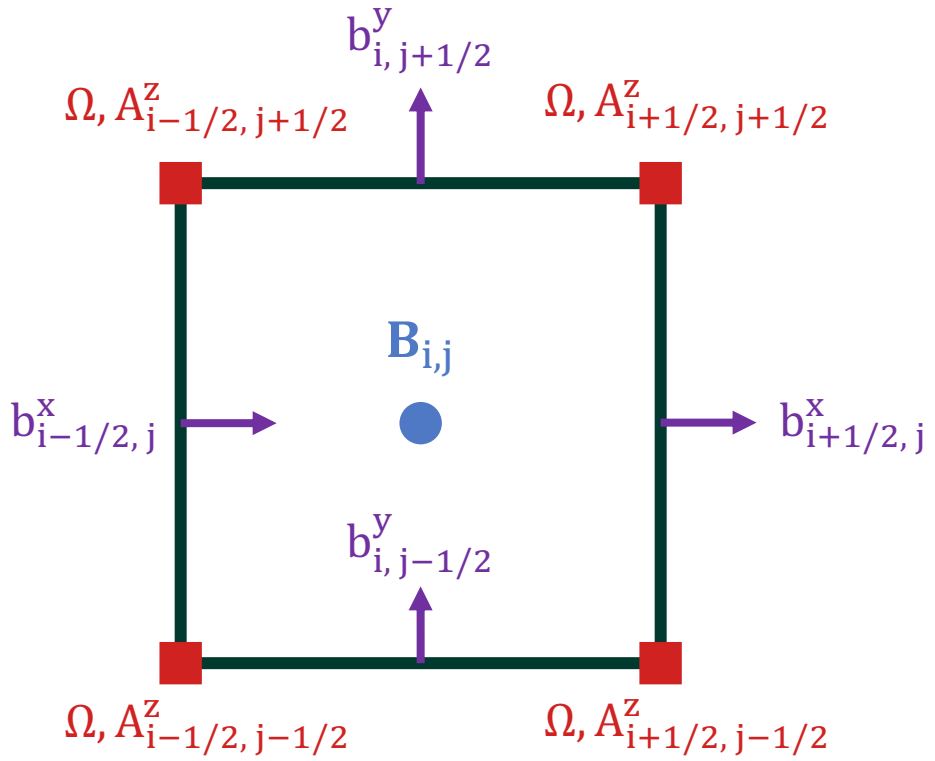
# Paper #2: Constrained Transport

- Start with the integral form of Faraday's Law

$$\frac{\mathrm{d}}{\mathrm{dt}} \int_{\mathrm{s}} \mathbf{B} \cdot \mathrm{d}\boldsymbol{A} = -\int_{\mathrm{s}} (\nabla \times \mathbf{E}) \cdot \mathrm{d}\boldsymbol{A}$$

- Now apply Stokes' theorem

$$\int_{\mathrm{s}} (\nabla \times \mathbf{E}) \cdot \mathrm{d}\boldsymbol{A} = \oint_{\partial \mathrm{s}} \mathbf{E} \cdot \mathrm{d}\boldsymbol{l}$$

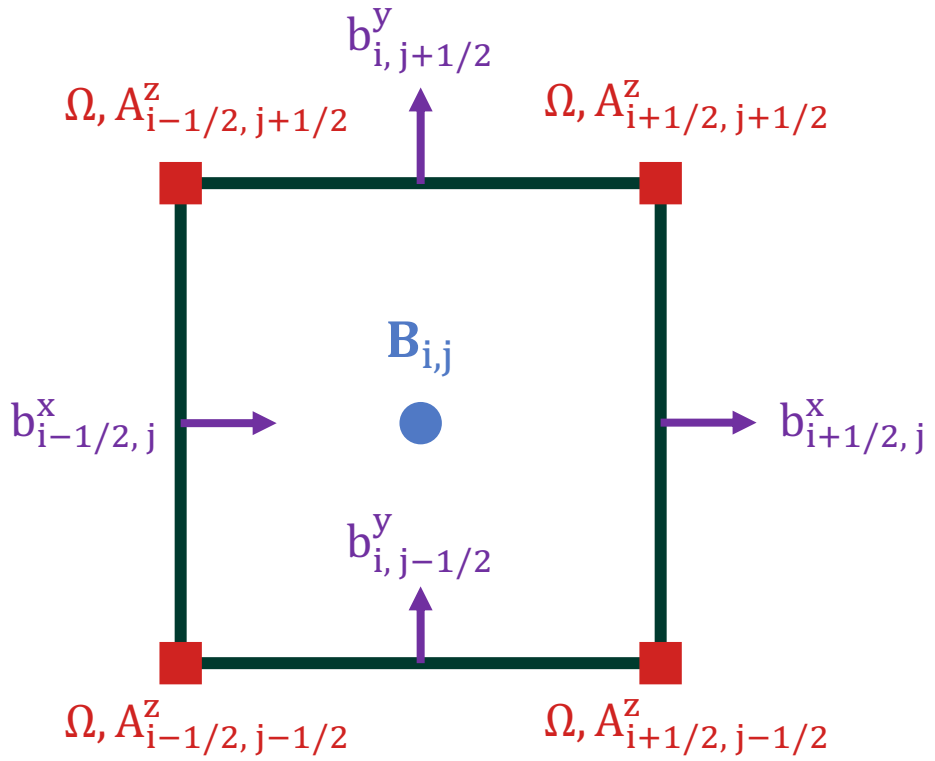


# Paper #2: Constrained Transport

- We end up with

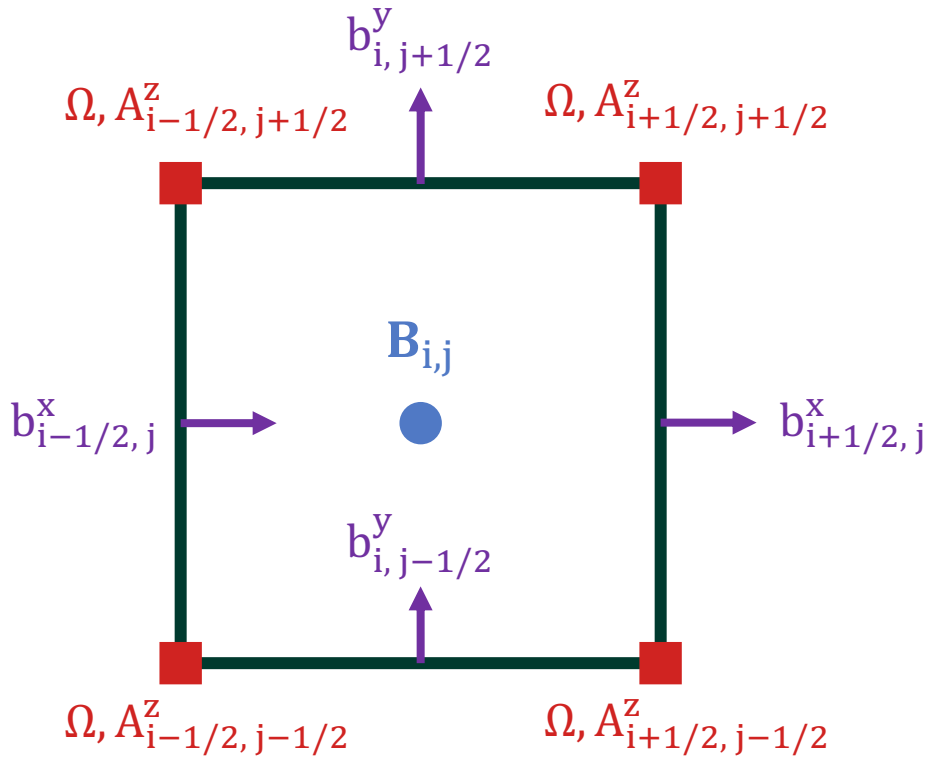
$$\frac{\mathrm{d}}{\mathrm{dt}} \int_{\mathrm{s}} \mathbf{B} \cdot \mathrm{d}\boldsymbol{A} = -\oint_{\partial \mathrm{s}} \mathbf{E} \cdot \mathrm{d}\boldsymbol{l}$$

- **B** is a surface quantity (flux through faces)
- **E** is a line quantity (circulation along edges)



# Paper #2: Constrained Transport

- $\nabla \cdot \mathbf{B} = \partial_x B_x + \partial_y B_y + \partial_z B_z = 0$



# Paper #2: Constrained Transport

*for 2D*

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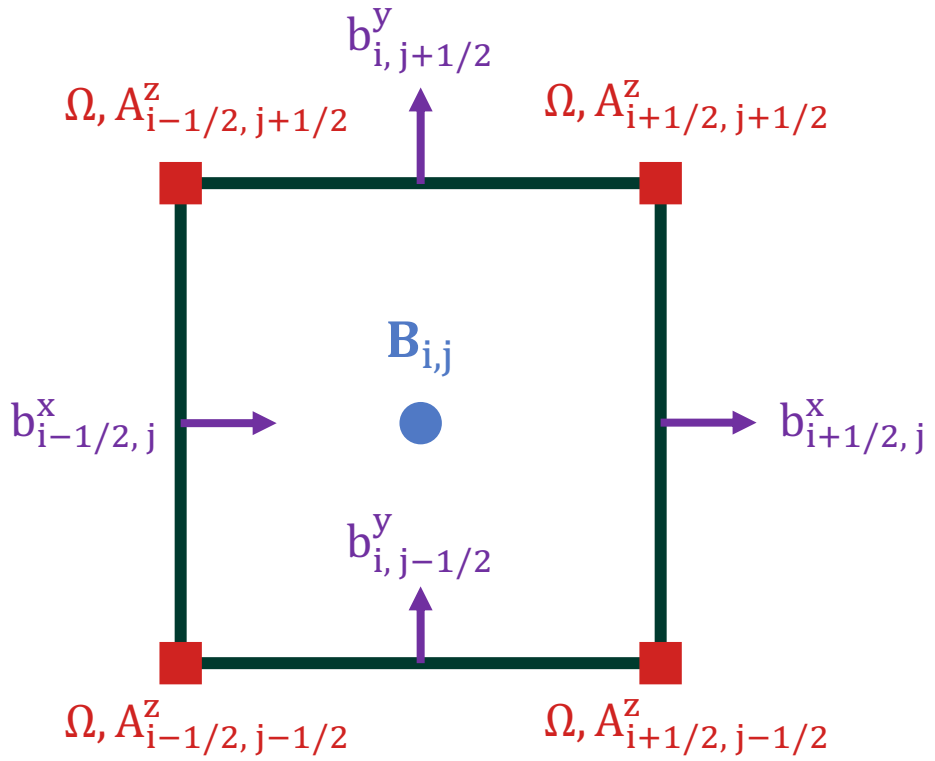
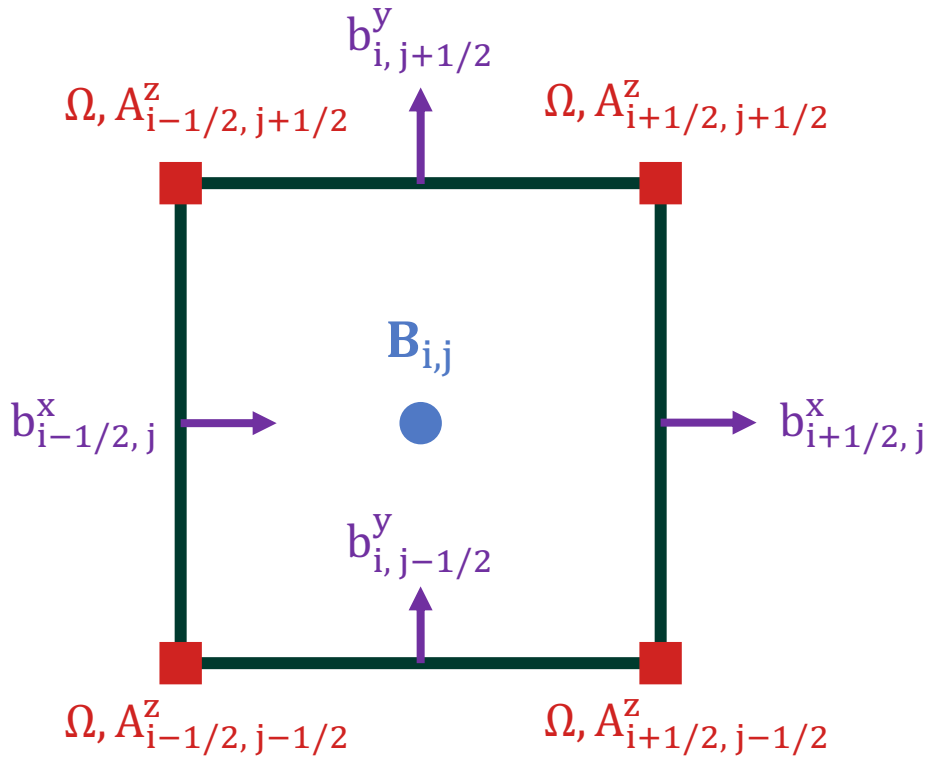


# Paper #2: Constrained Transport

*for 2D*

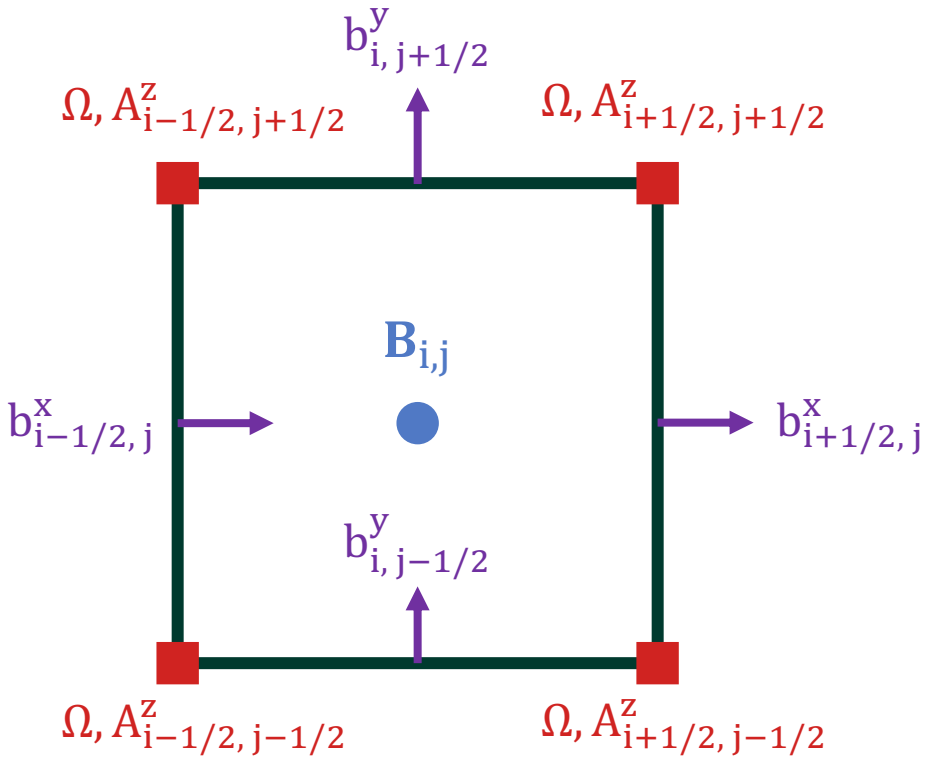
- $\nabla \cdot \mathbf{B} = \partial_x B_x + \partial_y B_y + \cancelto{0}{\partial_z B_z} = 0$

- $[\nabla \cdot \mathbf{B}]_{i,j} = \frac{1}{\Delta x}\left(b^x_{i+1/2,\,j} - b^x_{i-1/2,\,j}\right) + \frac{1}{\Delta y}\left(b^y_{i,\,j+1/2} - b^y_{i,\,j-1/2}\right)$



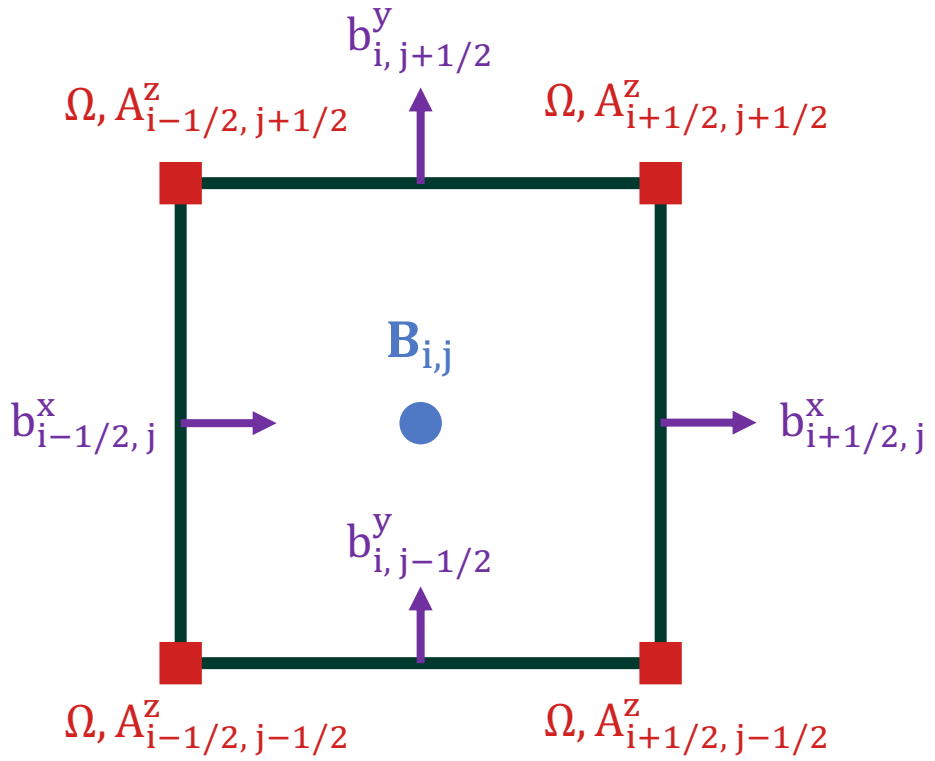
# Paper #2: Constrained Transport

- $\mathbf{B} = \nabla \times \mathbf{A}$
- In 2D
  - $b^x = \partial_y A^z$
  - $b^y = -\partial_x A^z$



# Paper #2: Constrained Transport

- $b^x_{i+1/2,\,j} = \frac{1}{\Delta y}\left(A^z_{i+1/2,\,j+1/2} - A^z_{i+1/2,\,j-1/2}\right)$
- $b^x_{i-1/2,\,j} = \frac{1}{\Delta y}\left(A^z_{i-1/2,\,j+1/2} - A^z_{i-1/2,\,j-1/2}\right)$
- $b^y_{i,\,j+1/2} = \frac{1}{\Delta x}\left(A^z_{i-1/2,\,j+1/2} - A^z_{i+1/2,\,j+1/2}\right)$
- $b^y_{i,\,j-1/2} = \frac{1}{\Delta x}\left(A^z_{i-1/2,\,j-1/2} - A^z_{i+1/2,\,j-1/2}\right)$

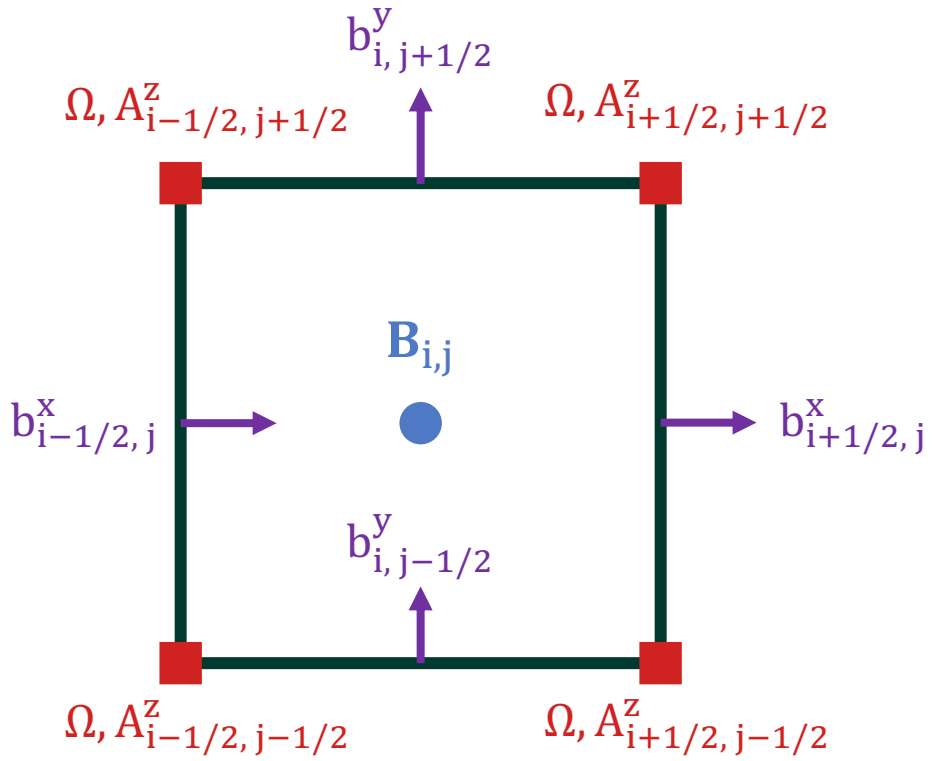


# Paper #2: Constrained Transport

If I plug those back into

$$[\nabla \cdot \mathbf{B}]_{i,j} = \frac{1}{\Delta x}\left(b^x_{i+1/2,\,j} - b^x_{i-1/2,\,j}\right) + \frac{1}{\Delta y}\left(b^y_{i,\,j+1/2} - b^y_{i,\,j-1/2}\right)$$

I will end up with $[\nabla \cdot \mathbf{B}]_{i,j} = 0$ exactly!



# Paper #2: Constrained Transport

$$[\nabla \cdot \mathbf{B}]_{i,j} = \frac{1}{\Delta x}\frac{1}{\Delta y}\left(\left(A^z_{i+1/2,\,j+1/2} - A^z_{i+1/2,\,j-1/2}\right) - \left(A^z_{i-1/2,\,j+1/2} - A^z_{i-1/2,\,j-1/2}\right)\right)$$

$$+ \frac{1}{\Delta y}\frac{1}{\Delta x}\left(\left(A^z_{i-1/2,\,j+1/2} - A^z_{i+1/2,\,j+1/2}\right) - \left(A^z_{i-1/2,\,j-1/2} - A^z_{i+1/2,\,j-1/2}\right)\right)$$

# Paper #2: Constrained Transport

**Step 0.** Primary quantities: $\rho^n,\ \rho\vec{u}^{\,n},\ \mathcal{E}^n,\ b_1^n,\ b_2^n,\ B_3^n,\ \mathbb{A}^n.$

**Step 1.** Average edge magnetic field:

$$\Longrightarrow [B_1]_{ij}^n = \frac{1}{2}\Big([b_1]_{i-1/2,j}^n + [b_1]_{i+1/2,j}^n\Big),$$
$$\Longrightarrow [B_2]_{ij}^n = \frac{1}{2}\Big([b_2]_{i,j-1/2}^n + [b_2]_{i,j+1/2}^n\Big).$$

**Step 2.** Solve the MHD equations using the base scheme:

$$\Longrightarrow \rho^{n+1},\ \rho\vec{u}^{\,n+1},\ \mathcal{E}^{n+1},\ \vec{B}^{\star}.$$

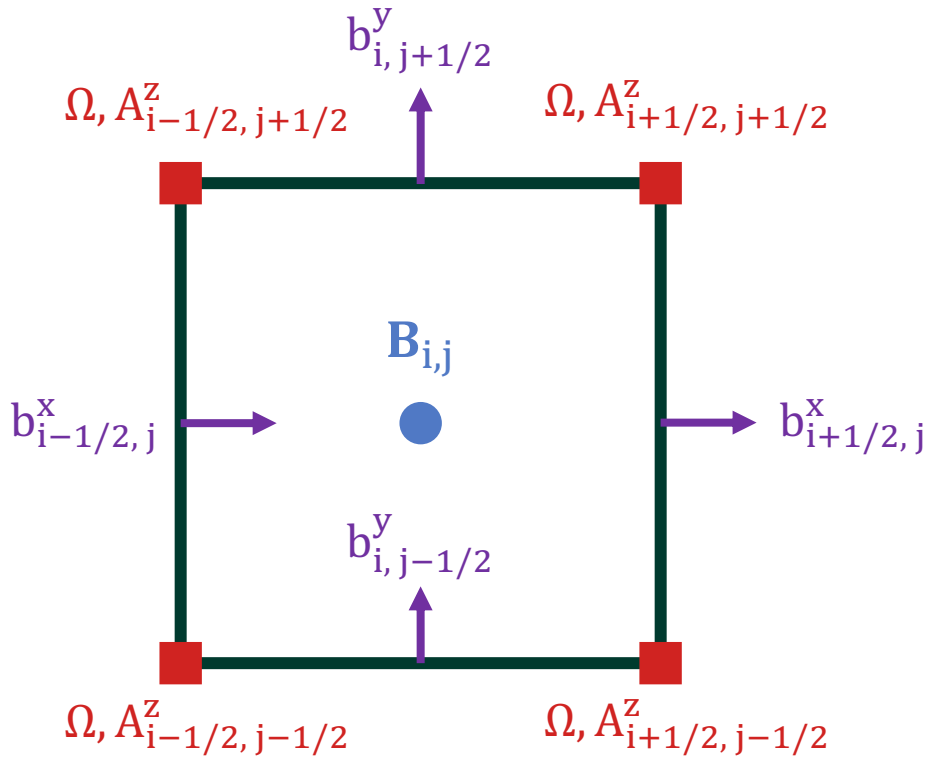
**Step 3.** Construct electric field at cell corners: $\Omega_{i-1/2,j-1/2}^{n+1/2}$.

**Step 4.** Update the magnetic potential:

$$\Longrightarrow \mathbb{A}_{i-1/2,j-1/2}^{n+1} = \mathbb{A}_{i-1/2,j-1/2}^{n} - \Delta t\,\Omega_{i-1/2,j-1/2}^{n+1/2}.$$

**Step 5.** Compute $\vec{\nabla}\times\vec{\mathbb{A}}$:

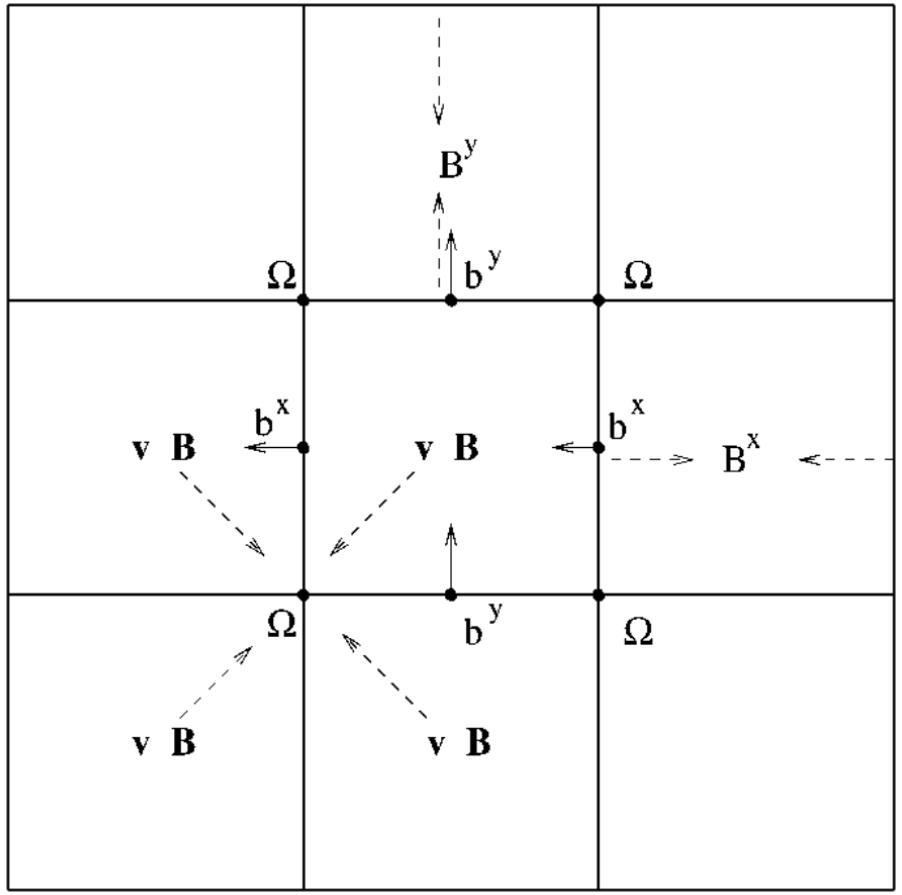
$$\Longrightarrow [b_1]_{i-1/2,j}^{n+1} = \frac{1}{\Delta y}\Big(\mathbb{A}_{i-1/2,j+1/2}^{n+1} - \mathbb{A}_{i-1/2,j-1/2}^{n+1}\Big),$$
$$\Longrightarrow [b_2]_{i,j-1/2}^{n+1} = \frac{1}{\Delta x}\Big(\mathbb{A}_{i-1/2,j-1/2}^{n+1} - \mathbb{A}_{i+1/2,j-1/2}^{n+1}\Big).$$



# Paper #2: Constrained Transport

Many ways to compute Step 3…

$$\Omega_{j+1/2,k+1/2} = -\bar{\boldsymbol{v}}^{n+1/2}_{j+1/2,k+1/2} \times \bar{\boldsymbol{B}}^{n+1/2}_{j+1/2,k+1/2}$$



# Paper #2: Constrained Transport

- Pros:
  - The magnetic field is exactly divergence free
- Cons:
  - Complex structure with edge, face, and cell centered values
  - Harder with complex geometries (unstructured meshes, curvilinear coordinates)

# Paper #3: Unstaggered Constrained Transport

An Unstaggered, High-Resolution Constrained Transport Method for Magnetohydrodynamic Flows

James A. Rossmamith - 2006

# Paper #3: Unstaggered Constrained Transport

- Because it is divergence-free, the magnetic field $\mathbf{B}$ can be written as the curl of a magnetic vector potential $\mathbf{A}$

$$\mathbf{B} = \nabla \times \mathbf{A}$$

- The Induction equation can be rewritten as

$$\frac{\partial \mathbf{A}}{\partial \mathrm{t}} + (\nabla \times \mathbf{A}) \times \mathbf{u} = 0$$

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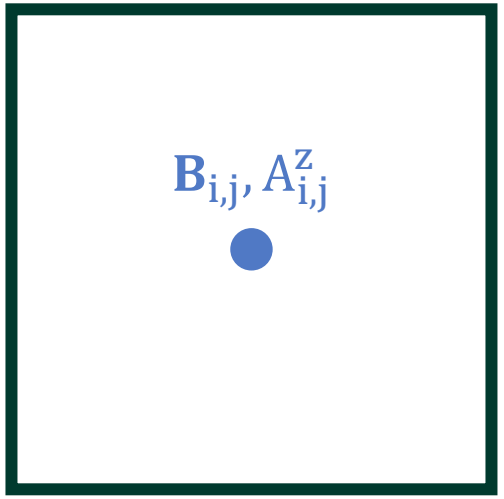
** This is the same as before: $\frac{\partial \mathbf{A}}{\partial \mathrm{t}} + \mathbf{\Omega} = 0$ where $\mathbf{\Omega} = \mathbf{B} \times \mathbf{u}$ **

# Paper #3: Unstaggered Constrained Transport

- At cell centers: $\rho_{i,j}$, $\rho\mathbf{u}_{i,j}$, $\mathcal{E}_{i,j}$, $\mathbf{B}_{i,j}$, $\mathbf{A}_{i,j}$



# Paper #3: Unstaggered Constrained Transport

**Step 0.** Primary quantities: $\rho^n,\ \rho\vec{u}^{\,n},\ \mathcal{E}^n,\ \vec{B}^{\,n},\ \mathbb{A}^n.$

**Step 1.** Solve the MHD equations using the base scheme:

$$\Longrightarrow \rho^{n+1},\ \rho\vec{u}^{\,n+1},\ \mathcal{E}^{n+1},\ \vec{B}^{\star}.$$

(NOTE: $\vec{B}^{\star}$ is discarded and need not be stored.)

**Step 2.** Time average the velocity field:

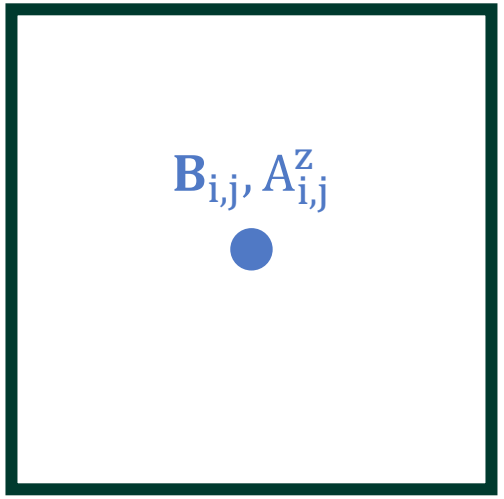
$$\Longrightarrow \vec{u}^{\,n+1/2} = \tfrac{1}{2}\left(\vec{u}^{\,n} + \vec{u}^{\,n+1}\right).$$

**Step 3.** Solve the magnetic potential equation (5.1) using $\vec{u}^{\,n+1/2}$: $\mathbb{A}^{n+1}$.

**Step 4.** Compute $\vec{\nabla} \times \vec{\mathbb{A}}$:

$$\Longrightarrow [B_1]_{ij}^{n+1} = \frac{1}{2\Delta y}\left(\mathbb{A}_{i,j+1}^{n+1} - \mathbb{A}_{i,j-1}^{n+1}\right),$$

$$\Longrightarrow [B_2]_{ij}^{n+1} = \frac{1}{2\Delta x}\left(\mathbb{A}_{i-1,j}^{n+1} - \mathbb{A}_{i+1,j}^{n+1}\right).$$



# Paper #3: Unstaggered Constrained Transport

- Pros:
  - The magnetic field is exactly divergence free
  - Only have cell centered values
- Cons:
  - Additional evolution equation for the magnetic vector potential
  - From personal experience, boundary conditions can be a pain for $\mathbf{A}$

# Paper #4: Projection Methods (Hodge Decomposition)

The Effect of Nonzero $\nabla \cdot \mathbf{B}$ on the Numerical Solution of the Magnetohydrodynamic Equations

Brackbill and Barnes - 1980

# Paper #4: Projection Methods (Hodge Decomposition)

- Returning to Helmholtz Decomposition Theorem…

$$\mathbf{B}^* = \nabla\phi + \nabla \times \mathbf{A}$$

- The physically meaningful part of $\mathbf{B}^*$ comes from $\nabla \times \mathbf{A}$

# Paper #4: Projection Methods (Hodge Decomposition)

- Returning to Helmholtz Decomposition Theorem…

$$\mathbf{B}^* = \nabla\phi + \nabla \times \mathbf{A}$$

- The physically meaningful part of $\mathbf{B}^*$ comes from $\nabla \times \mathbf{A}$

- Taking the divergence of both sides, we obtain a Poisson equation

$$\nabla \cdot \mathbf{B}^* = \nabla^2\phi$$

# Paper #4: Projection Methods (Hodge Decomposition)

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$$\mathbf{B}^{n+1} = (\nabla\phi + \nabla \times \mathbf{A}) - \nabla\phi$$

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$$\mathbf{B}^{n+1} = \nabla \times \mathbf{A}$$

# Paper #4: Projection Methods (Hodge Decomposition)

- Pros:
  - The magnetic field is exactly divergence free
- Cons:
  - Solving the Poisson equation $\nabla \cdot \mathbf{B}^* = \nabla^2 \phi$ is computationally expensive

# Paper #5: Hyperbolic Divergence Cleaning

Hyperbolic Divergence Cleaning for the MHD Equations

A. Dedner, F. Kemm, D. Kröner, C.-D., Munz, T. Schnitzer, and M. Wesenberg - 2002

# Paper #5: Hyperbolic Divergence Cleaning

- Based on a divergence correction technique for Maxwell's Equations

# Paper #5: Hyperbolic Divergence Cleaning

- Based on a divergence correction technique for Maxwell's Equations
- Add a new unknown function $\psi$ to both the Induction and Involution equations

$$\partial_t \mathbf{B} + \nabla \cdot (\mathbf{u} \otimes \mathbf{B} - \mathbf{B} \otimes \mathbf{u}) + \nabla\psi = 0$$

$$\mathcal{D}(\psi) + \nabla \cdot \mathbf{B} = 0$$

where $\mathcal{D}$ is a linear differential operator

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- **Goal**: Introduce $\psi$ to couple the divergence constraint to the hyperbolic system

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- They found a mixed parabolic, hyperbolic correction term worked best

$$\mathcal{D}(\psi) = \frac{1}{c_h^2}\partial_t\psi + \frac{1}{c_p^2}\psi \text{ with } c_h \in (0,\infty) \text{ and } c_p \in (0,\infty)$$

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- Local divergence errors are dissipated/smoothed out ($c_p$) and propagated to the boundary with the finite speed $c_h > 0$

# Paper #5: Hyperbolic Divergence Cleaning

$$
\begin{aligned}
\partial_t \rho + \partial_x(\rho u_x) &= 0, \\
\partial_t(\rho u_x) + \partial_x\left(\rho u_x^2 + p + \frac{1}{2}\left(B_y^2 + B_z^2 - B_x^2\right)\right) &= 0, \\
\partial_t(\rho u_y) + \partial_x(\rho u_x u_y - B_x B_y) &= 0, \\
\partial_t(\rho u_z) + \partial_x(\rho u_x u_z - B_x B_z) &= 0, \\
\partial_t B_x + \partial_x \psi &= 0, \\
\partial_t B_y + \partial_x(B_y u_x - B_x u_y) &= 0, \\
\partial_t B_z + \partial_x(B_z u_x - B_x u_z) &= 0, \\
\partial_t e + \partial_x\left[u_x\left(e + p + \frac{1}{2}\mathbf{B}^2\right) - B_x(u_x B_x + u_y B_y + u_z B_z)\right] &= 0, \\
\partial_t \psi + c_h^2 \partial_x B_x &= -\frac{c_h^2}{c_p^2}\psi.
\end{aligned}
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\boxed{\partial_t B_x + \partial_x \psi = 0,} \\
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\end{aligned}
$$

# Paper #5: Hyperbolic Divergence Cleaning

- Given the primitive variables: $\mathbf{W} = (\rho, u_x, u_y, u_z, B_x, B_y, B_z, p, \psi)^T$

- The Jacobian of the system is given as:

$$\mathcal{A}(\mathbf{W}) = \begin{pmatrix} u_x & \rho & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & u_x & 0 & 0 & -\frac{B_x}{\rho} & \frac{B_y}{\rho} & \frac{B_z}{\rho} & \frac{1}{\rho} & 0 \\ 0 & 0 & u_x & 0 & -\frac{B_y}{\rho} & -\frac{B_x}{\rho} & 0 & 0 & 0 \\ 0 & 0 & 0 & u_x & -\frac{B_z}{\rho} & 0 & -\frac{B_x}{\rho} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & B_y & -B_x & 0 & -u_y & u_x & 0 & 0 & 0 \\ 0 & B_z & 0 & -B_x & -u_z & 0 & u_x & 0 & 0 \\ 0 & \gamma p & 0 & 0 & (\gamma - 1)\mathbf{u} \cdot \mathbf{B} & 0 & 0 & u_x & (1-\gamma)B_x \\ 0 & 0 & 0 & 0 & c_h^2 & 0 & 0 & 0 & 0 \end{pmatrix}$$

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Jacobian is identical to Powell before correction

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# Paper #5: Hyperbolic Divergence Cleaning

- The eigenvalues are given as:

$$\lambda_1 = -c_h, \quad \lambda_2 = u_x - c_f, \quad \lambda_3 = u_x - c_a, \quad \lambda_4 = u_x - c_s, \quad \lambda_5 = u_x,$$
$$\lambda_6 = u_x + c_s, \quad \lambda_7 = u_x + c_a, \quad \lambda_8 = u_x + c_f, \quad \lambda_9 = c_h.$$

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- The wave speeds $c_f$, $c_s$, $c_a$ are the (unchanged) fast, slow, and Alfvén waves
- $c_h$ is the speed at which local divergence errors are propagated to the boundary
- Only the waves traveling with speeds $\lambda_1$ and $\lambda_9$ can carry a change in $B_x$ or $\psi$

# Paper #5: Hyperbolic Divergence Cleaning

- Pros:
  - Easy to add, works with cell centered magnetic fields
- Cons:
  - Does not remove divergence errors, only damps/transports them away
  - Includes tuning parameters ($c_h$ and $c_p$)